

Comparison of Means across Several Group with R

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1 Loading Data

We will use the *Palmer Penguin* data set available directly from the `palmerpenguins` package. You can learn more about the data at allisonhorst.github.io/palmerpenguins. We will explore how penguin body mass varies across conditions, starting with species.

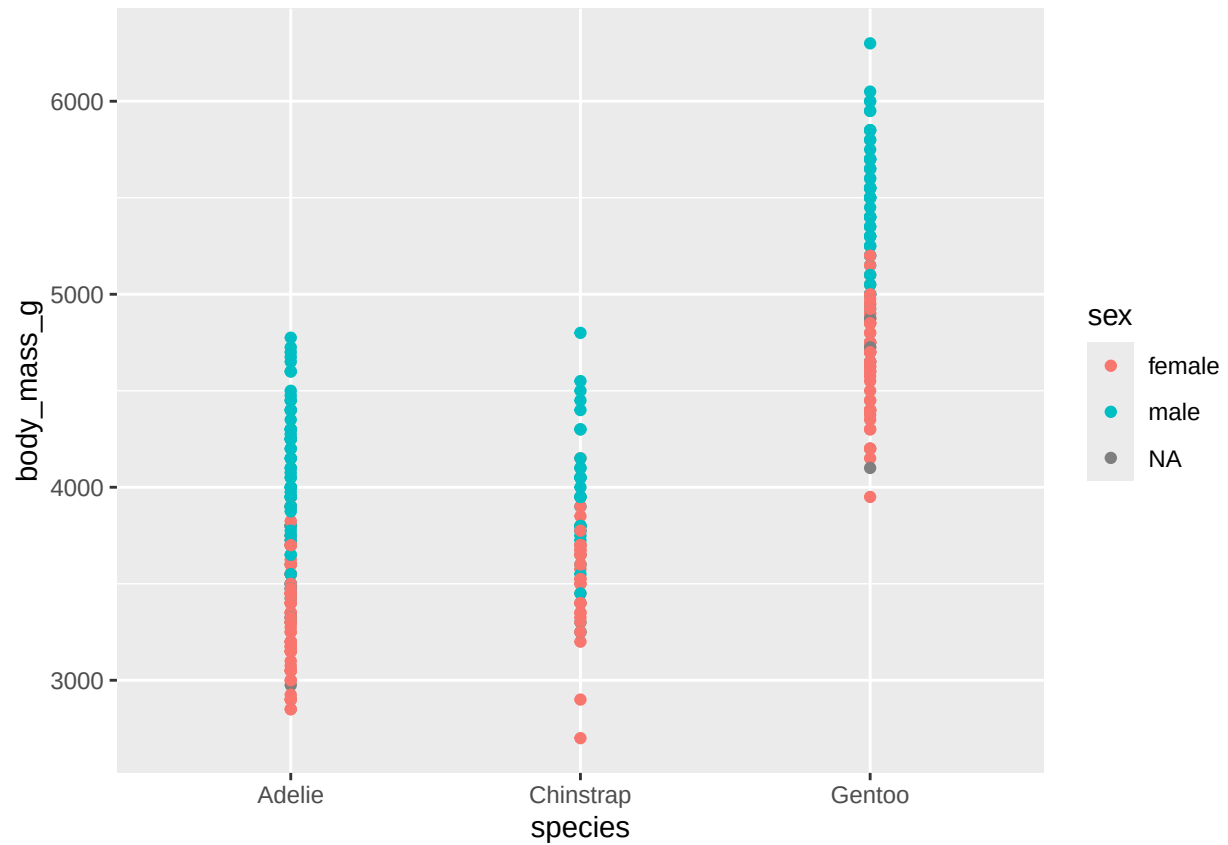
```
# install.packages("palmerpenguins")
library(palmerpenguins)
library(ggplot2)
library(dplyr)

dat <- as.data.frame(penguins) # rename dataset
str(dat) # structure of dataset
```

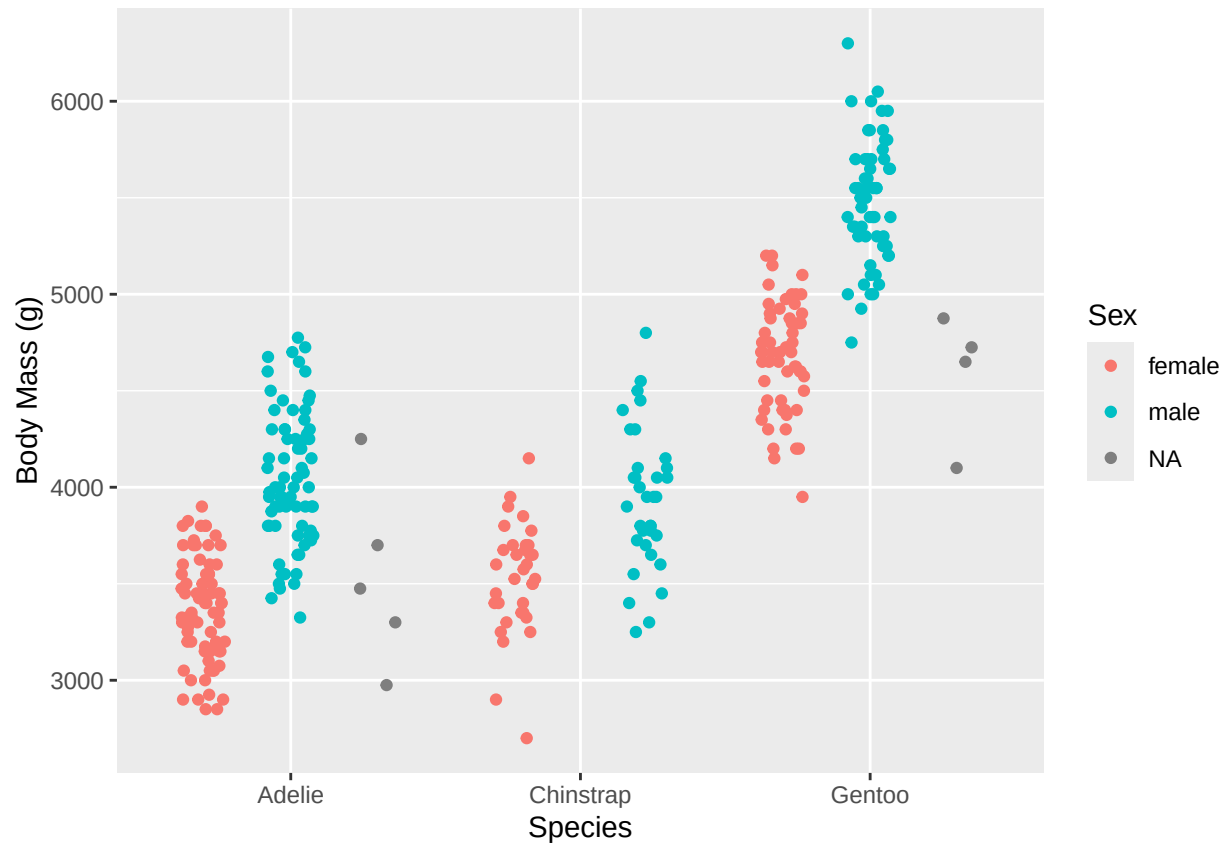
2 Exploring the data

2.1 Plot

```
ggplot(dat, aes(x = species, y = body_mass_g, color = sex)) +
  geom_point()
```



```
# Improve readability  
  
ggplot(dat, aes(x = species, y = body_mass_g, color = sex)) +  
  geom_point(position=position_jitterdodge(dodge.width=0.9)) +  
  ylab("Body Mass (g)") +  
  xlab("Species") +  
  scale_color_discrete(name = "Sex")
```



2.2 Subset

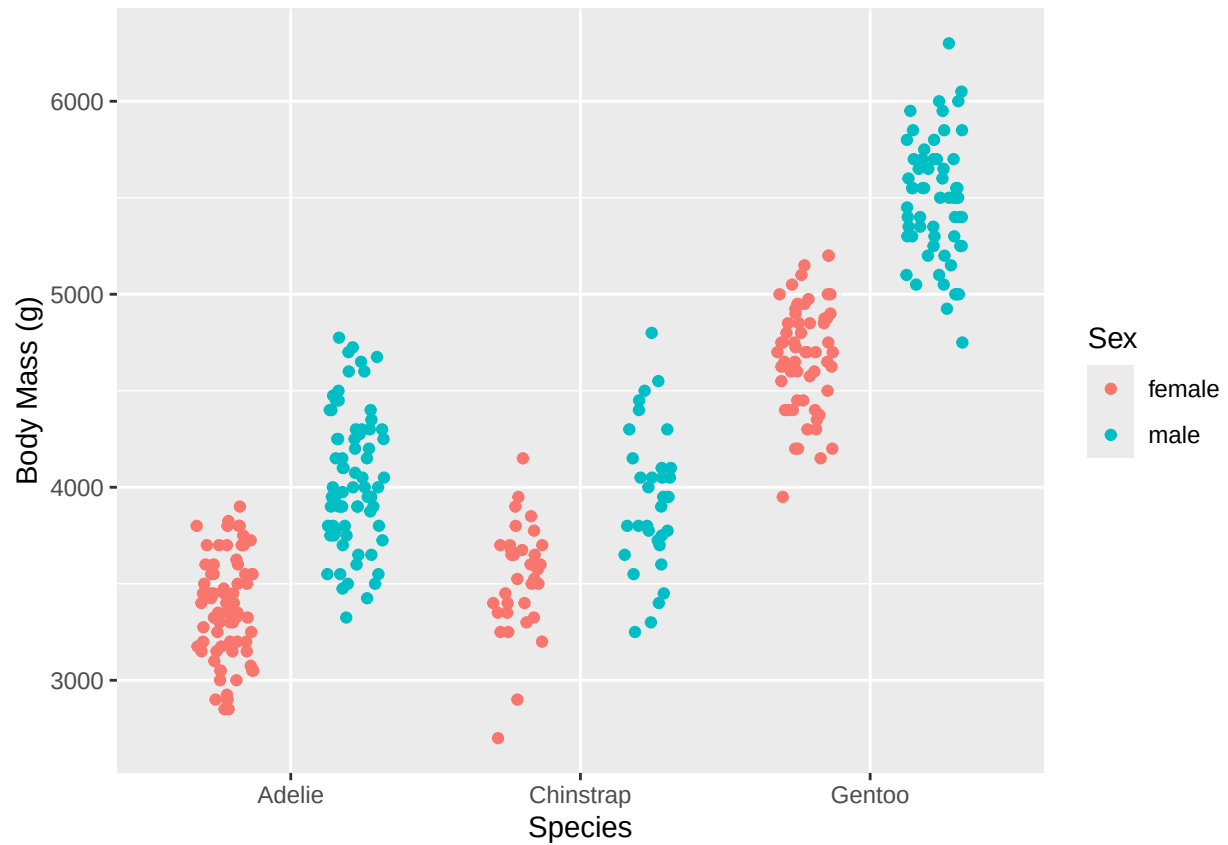
Focus on observations with sex recorded

```
dat = subset(dat, !is.na(sex))
```

```
unique(dat$sex)
```

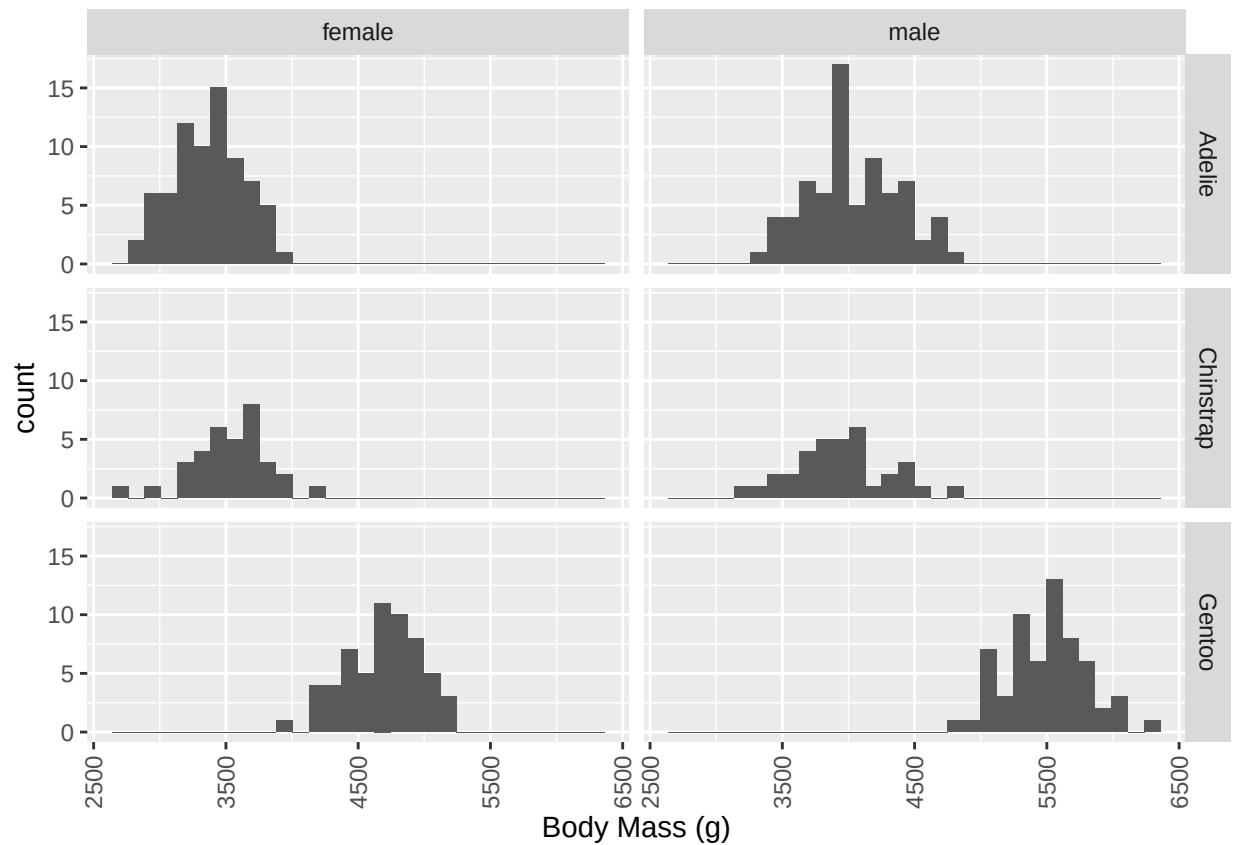
2.3 Re-plot

```
ggplot(dat, aes(x = species, y = body_mass_g, color = sex)) +  
  geom_point(position=position_jitterdodge(dodge.width=0.9)) +  
  ylab("Body Mass (g)") +  
  xlab("Species") +  
  scale_color_discrete(name = "Sex")
```



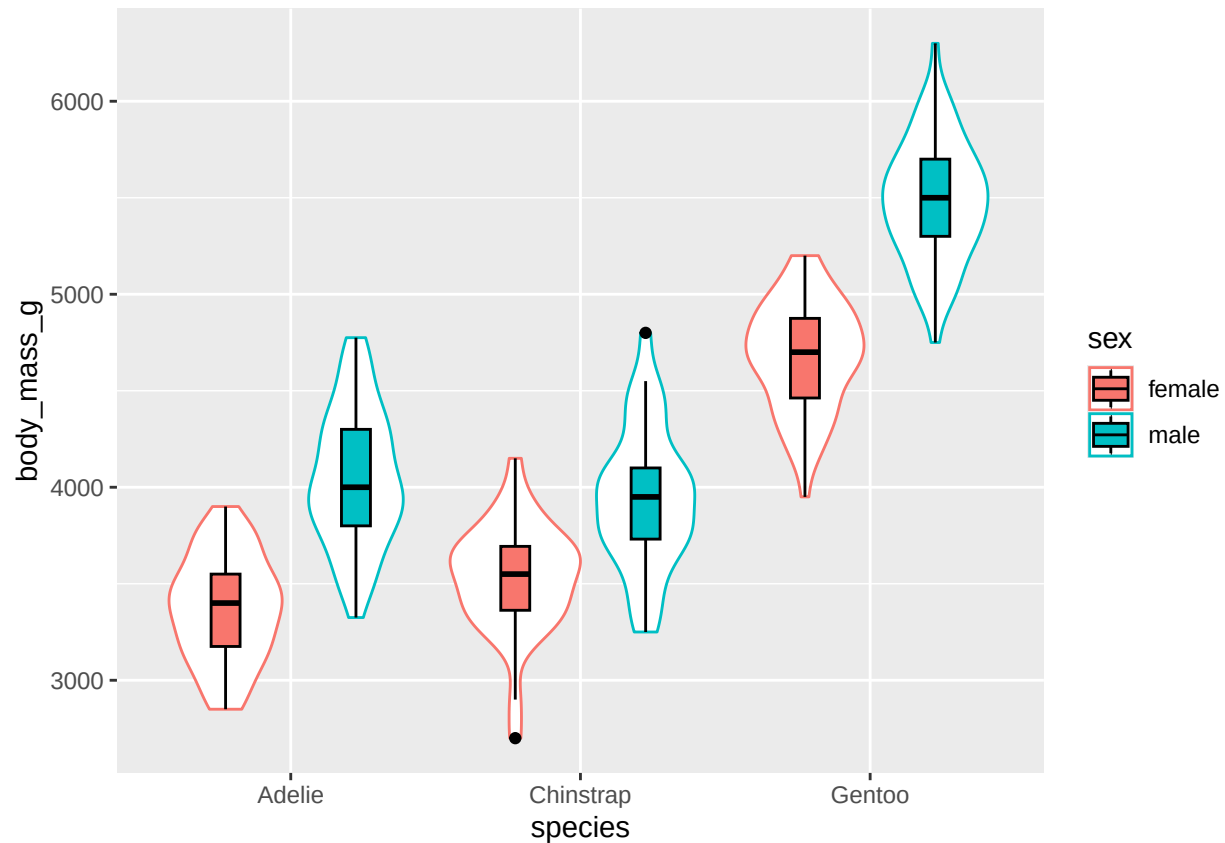
```
# Display distributions
```

```
ggplot(dat, aes(x = body_mass_g)) +  
  geom_histogram() +  
  xlab("Body Mass (g)") +  
  facet_grid(species ~ sex) +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Other option

```
ggplot(dat) +
  aes(x = species, y = body_mass_g) +
  geom_violin(aes(color = sex), position = position_dodge(0.9)) +
  geom_boxplot(color = "black", aes(fill = sex), width = 0.2, position = position_dodge(0.9))
```



Discuss distribution shapes, and differences (or lack of).

2.4 Summarize

```
# mean by group
group_by(dat, sex, species) %>%
  summarise(
    mean = round(mean(body_mass_g, na.rm = TRUE)),
    sd = round(sd(body_mass_g, na.rm = TRUE))
  )
```

3 Statistical Test

Let's consider one factor at a time, and consider males only. Here we aim at comparing a continuous variable (body mass) across 3 groups (species). An ANOVA (if assumptions are verified) or a Kruskal-Wallis test would be appropriate.

```
dat_male = subset(dat, sex == "male")
```

3.1 Assumptions

3.1.1 Independance

Observations from each population are independent of one another. This is likely the case (would be determined by the study design).

3.1.2 Normality

Data is normally distributed within each group. This is visible on the histogram. Make sure the data is stratified by group.

This can also be verified with the Shapiro-Wilk test, **for each group**.

```
shapiro.test(dat_male[dat_male$species == "Adelie", "body_mass_g"])

##
## Shapiro-Wilk normality test
##
## data:  dat_male[dat_male$species == "Adelie", "body_mass_g"]
## W = 0.98269, p-value = 0.416

shapiro.test(dat_male[dat_male$species == "Chinstrap", "body_mass_g"])

##
## Shapiro-Wilk normality test
##
## data:  dat_male[dat_male$species == "Chinstrap", "body_mass_g"]
## W = 0.98414, p-value = 0.891

shapiro.test(dat_male[dat_male$species == "Gentoo", "body_mass_g"])

##
## Shapiro-Wilk normality test
##
## data:  dat_male[dat_male$species == "Gentoo", "body_mass_g"]
## W = 0.99336, p-value = 0.985
```

The observed distributions do not statistically differ from a normal distribution.

3.1.3 Homoscedasticity

Variances are equal across populations. This is visible on the histogram.

This can also be verified with the variance F test test.

```
bartlett.test(body_mass_g ~ species, dat_male)

##
## Bartlett test of homogeneity of variances
##
## data:  body_mass_g by species
## Bartlett's K-squared = 1.081, df = 2, p-value = 0.5824
```

The variances are not significantly different from one another.

3.2 Test

```
# Fit model
mod = aov(body_mass_g ~ species,
          data = dat_male)

# Return outputs (including p-value)
summary(mod)

##              Df    Sum Sq Mean Sq F value Pr(>F)
## species      2 84728125 42364062   370.4 <2e-16 ***
```

```
## Residuals    165 18871872    114375
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We reject the null hypothesis, i.e., we detect a difference. We cannot tell which group differs from which though.

3.3 Post-hoc Test

We need to run pair-wise tests with an adjustment for multiple testing to identify which groups differs from which.

Based on the assumptions we verified above, a t-test is appropriate.

```
pairwise.t.test(dat_male$body_mass_g, dat_male$species)
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data:  dat_male$body_mass_g and dat_male$species
##
##           Adelie Chinstrap
## Chinstrap 0.14      -
## Gentoo    <2e-16 <2e-16
##
## P value adjustment method: holm
```

Gentoos differ from the two others.

3.4 Explanation

Based on an ANOVA, followed by pair-wise t-tests, male gentoo penguins are significantly heavier than the other species, with gentoo penguins weighting on average (standard deviation) 5,485g (313g) vs 4,043g (347g) for Adelie penguins, and 3,939g (362g) for chinstrap penguins.

4 Non-parametric Equivalent

A non-parametric test would also work (they almost always work!). For a continuous variable recorded across several groups, Kruskal-Wallis rank test is appropriate.

```
# Kruskal-Wallis test
kruskal.test(body_mass_g ~ species, dat_male)
```

```
##
## Kruskal-Wallis rank sum test
##
## data:  body_mass_g by species
## Kruskal-Wallis chi-squared = 116.5, df = 2, p-value < 2.2e-16
```

Followed by pair-wise Wilcoxon rank test.

```
# Kruskal-Wallis test
pairwise.wilcox.test(dat_male$body_mass_g, dat_male$species)
```

```
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  dat_male$body_mass_g and dat_male$species
```



```
##
##           Adelie  Chinstrap
## Chinstrap 0.19    -
## Gentoo    < 2e-16 1.7e-15
##
## P value adjustment method: holm
```

We reach the same conclusion.

5 Side note: two-way ANOVAs

We could run a two-way ANOVA to account for sex and species at the same time.

```
# Fit model
mod = aov(body_mass_g ~ species + sex,
          data = dat)

# Return outputs (including p-value)
summary(mod)
```

```
##           Df      Sum Sq  Mean Sq F value Pr(>F)
## species      2 145190219 72595110   724.2 <2e-16 ***
## sex          1  37090262 37090262   370.0 <2e-16 ***
## Residuals    329  32979185   100241
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This kind of multivariate models is easier to run and interpret with regression approaches though.